Tide retrospective: asymptotic division

quotients of order-N expansions

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Abstract

Stage 6 of the forward-expansion programme, the generic division lemma the design consult prescribed: if f and g admit order-N asymptotic polynomials at 0^+ with coefficients a, b and $b_0 \neq 0$, then f/g admits the order-N expansion with the division coefficients $c_0 = a_0/b_0$, $c_j = \left(a_j - \sum_{i=1}^{j} b_i c_{j-i}\right)/b_0$. The proof telescopes $f - g\,C$ into the two expansion errors plus the polynomial $A - B \cdot C$, whose coefficients vanish through degree N by the convolution identity — verified through the Cauchy product of coefficient polynomials, the same `Polynomial.coeff_mul` mechanism as stages 4–5a — so that it carries q^{N+1}; the quotient then divides by g, eventually bounded below by $|b_0|/2$ since $g \to b_0$.

1 Setting

The moment expansion is the quotient of two instances of the numerator expansion (the partition function is the observable-one case), so this is the last infrastructure piece before the public theorems. The lemma is stated at the same generic level as stage A1's expansion predicate: no domain, no measure, just functions of the scale.

2 The thing formalised

`divisionCoeff` by strong recursion (through `Finset.attach` with a `termination_by` on the index; the unattached recursion equation recovered by `Finset.sum_attach`); the convolution identity `divisionCoeff_conv`; the coefficient polynomial `coeffPoly` with evaluation, coefficient, and degree lemmas; the Cauchy product through degree N (`coeffPoly_mul_coeff`, via `Finset.Nat.sum_antidiagonal_eq_sum_range_succ_mk`); the vanishing-tail bound; and `isAsymptoticExpansionTo_div`. The eventual lower bound on the denominator comes from transporting the expansion error to a limit (`isLittleO_one_iff` after comparing q^N with the constant function) and `Tendsto.eventually_const_le` at $|b_0|/2$. A numerical check of both the convolution identity and the $o(q^N)$ decay preceded the Lean.

3 Where progress stalled

Two small rounds: `div_eq_iff` pattern-matches division on the left of the equation, so an equation with the quotient on the right wants `eq_div_iff`; and an untyped `have :=`

`(h.tendsto 0).mono_left nhdsWithin_le_nhds` cannot synthesize the within-set — the identical term elaborates against an explicitly typed `have`.

4 Mathlib lookup versus new mathematics

Lookup: `Finset.sum_attach`, `Finset.Nat.sum_antidiagonal_eq_sum_range_succ_mk`, `IsLittleO.mul_isBigO`, `Tendsto.isBigO_one`, `isLittleO_one_iff`, `eq_div_iff`, `Tendsto.eventually_const_le`. New: nothing — formal power series division specialized to a finite order, with the collection mechanism this programme has now used three times.

5 What the next tide inherits

Stage 7, the last: positivity of the partition function's constant coefficient ($a_0(\mathbf{1}) = \int e^{-T_2} > 0$, from `correctionCoeffFn_zero` and the positivity of Gaussian integrals), the moment expansion (`rescaledMoment` = numerator / partition through this tide's division), and the public existence and jet-comparison theorems of the germbij forward direction.